Iterative Methods in Linear Algebra - Part 2

Alexandra A. Gomes

March 4, 2026

Stationary Methods

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

- let $\mathbf{A} = \mathbf{M} - \mathbf{N}$ so that $\mathbf{A}\mathbf{x} = \mathbf{b}$ becomes

$$\begin{aligned} \mathbf{M}\mathbf{x} &= \mathbf{N}\mathbf{x} + \mathbf{b} \\ &= (\mathbf{M} - \mathbf{A})\mathbf{x} + \mathbf{b} \\ \mathbf{x} &= (\mathbf{I} - \mathbf{M}^{-1}\mathbf{A})\mathbf{x} + \mathbf{M}^{-1}\mathbf{b}; \end{aligned}$$

- leads to an iteration of the form

$$\mathbf{x}^{(\mathbf{k}+1)} = \mathbf{x}^{(\mathbf{k})} + \mathbf{M}^{-1}\left(\mathbf{b} - \mathbf{A}\mathbf{x}^{(\mathbf{k})}\right) = \mathbf{x}^{(\mathbf{k})} + \mathbf{M}^{-1}\mathbf{r}^{(\mathbf{k})};$$

- called *stationary* because iteration matrix does not change;
- the hard part is how to do the splitting of $\mathbf{A}$ into $\mathbf{M}$ and $\mathbf{N}$;

Stationary Methods: Choosing the Splitting Matrix $\mathbf{M}$

$$\mathbf{x}^{(\mathbf{k}+\mathbf{1})} = \mathbf{x}^{(\mathbf{k})} + \mathbf{M}^{-\mathbf{1}}\mathbf{r}^{(\mathbf{k})}$$

- $\mathbf{M}$ should be easy to "invert" (we never actually "invert"; we solve $\mathbf{M}\left(\mathbf{x}^{(\mathbf{k}+\mathbf{1})} - \mathbf{x}^{(\mathbf{k})}\right) = \mathbf{r}^{(\mathbf{k})}$);
- from $\mathbf{x} = (\mathbf{I} - \mathbf{M}^{-\mathbf{1}}\mathbf{A})\mathbf{x} + \mathbf{M}^{-\mathbf{1}}\mathbf{b}$, $\mathbf{M}$ should approximate $\mathbf{A}$.

Jacobi Method

$$\mathbf{x}^{(\mathbf{k}+\mathbf{1})} = \mathbf{x}^{(\mathbf{k})} + \mathbf{D}^{-\mathbf{1}}\mathbf{r}^{(\mathbf{k})}$$

- $\mathbf{M} = \mathbf{D} = \mathrm{diag}\left(a_{11}, \ldots, a_{nn}\right)$;
- $\mathbf{D}^{-\mathbf{1}}$ is trivial: reciprocals of the entries;
- the update is obtained by simply dividing $r_i^{(k)}$ by a_{ii}, that is, scaling the residual with the diagonal entry.

Jacobi Method

If $a_{ii} \neq 0$, for $i = 1, \ldots n$, Jacobi Method is

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left[b_i - \sum_{j=1, j \neq i}^{n} a_{ij} x_j^{(k)} \right],$$

- $x_i^{(k)}$ is adjusted to mitigate the i-th component of the residual;
- each $x_i^{(k)}$ can be computed independently, so parallelization is easy;
- if $a_{ii} = 0$ and the system is non-singular, then reorder the equations;
- rearrange the equations to get the largest a_{ii} in order to speed up the convergence of the algorithm.

Gauss-Seidel Method

$$\mathbf{x}^{(\mathbf{k}+\mathbf{1})} = \mathbf{x}^{(\mathbf{k})} + \mathbf{E}^{-\mathbf{1}}\mathbf{r}^{(\mathbf{k})}$$

- $\mathbf{M} = \mathbf{E}$, where $\mathbf{E}$ is the lower triangular part of $\mathbf{A}$ (including the diagonal);
- $\mathbf{E}$ is easy to "invert": $\mathbf{E}^{-\mathbf{1}}\mathbf{r}^{(\mathbf{k})}$ is a forward substitution pass to solve $\mathbf{E}\left(\mathbf{x}^{(\mathbf{k}+\mathbf{1})} - \mathbf{x}^{(\mathbf{k})}\right) = \mathbf{r}^{(\mathbf{k})}$;
- converges faster than Jacobi (when they converge!);
- but cannot be done in parallel.

Gauss-Seidel Method

Recall Jacobi Method: for $i = 1, \ldots n$,

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left[b_i - \sum_{j=1, j \neq i}^{n} a_{ij} x_j^{(k)} \right],$$

For Gauss-Seidel Method:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left[b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)} \right].$$

- the new components for $\mathbf{x}^{(\mathbf{k}+\mathbf{1})}$ are used; the idea is that we hope $|x_i^{(k+1)} - x_i| < |x_i^{(k)} - x_i|$.

Successive Over-Relaxation (SOR) Method

$$\mathbf{x}^{(\mathbf{k}+\mathbf{1})} = \omega \mathbf{x}_{\mathbf{GS}}^{(\mathbf{k}+\mathbf{1})} + (1-\omega)\mathbf{x}^{(\mathbf{k})}$$

- modification of Gauss-Seidel: ω-Gauss-Seidel;
- for $\omega = 1$, we get back Gauss-Seidel;
- faster convergence can be obtained with carefully chosen, problem dependent, $1 \leq \omega \leq 2$;
- $\mathbf{M}$ for SOR can be written as:

$$\mathbf{M} = \frac{1-\omega}{\omega}\mathbf{D} + \mathbf{E}.$$

Example

Solve

$$\begin{array}{rcrcrcrcr} 10x_1 & - & x_2 & + & 2x_3 & & & = & 6 \\ -x_1 & + & 11x_2 & - & x_3 & + & 3x_4 & = & 25 \\ 2x_1 & - & x_2 & + & 10x_3 & - & x_4 & = & -11 \\ & + & 3x_2 & - & x_3 & + & 8x_4 & = & 15 \end{array}$$

with initial vector $\mathbf{0}$ and convergence criterion

$$\frac{\left\|\mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{k-1})}\right\|_\infty}{\left\|\mathbf{x}^{(\mathbf{k})}\right\|_\infty} < 0.001.$$

The unique solution is $(1, 2, -1, 1)^T$.

Result

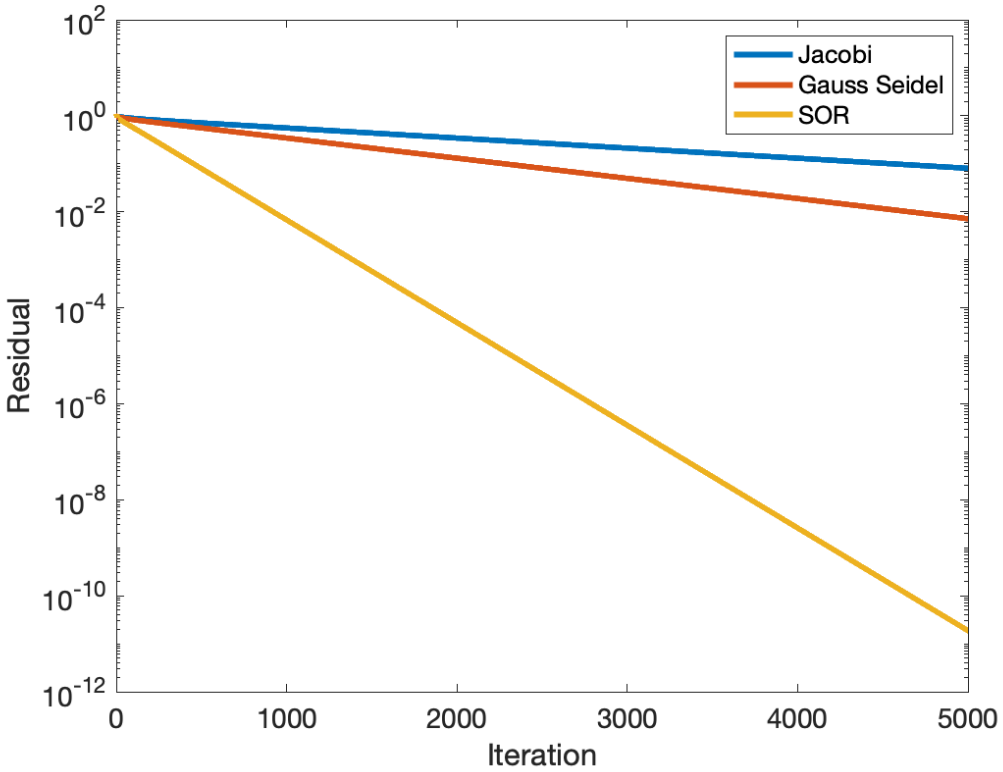
- with Jacobi's method:

k	2	5	7	10
$x_1^{(k)}$	1.0473	0.9890	0.9981	1.0001
$x_2^{(k)}$	1.7159	2.0114	2.0023	1.9998
$x_3^{(k)}$	-0.8052	-1.0103	-1.0020	-0.9998
$x_4^{(k)}$	0.8852	1.0214	1.0036	0.9998

- with Gauss-Siedel method:

k	2	5
$x_1^{(k)}$	1.030	1.0001
$x_2^{(k)}$	2.037	2.0000
$x_3^{(k)}$	-1.014	-1.0000
$x_4^{(k)}$	0.9844	1.0000

Comparison of Jacobi, Gauss-Seidel, and SOR Methods

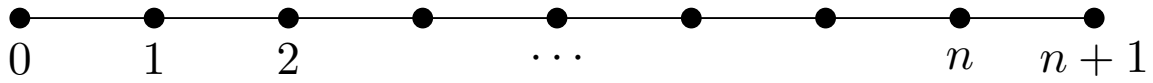


Discretization of an ODE

$$-\frac{d^2u}{dx^2} = b(x), \qquad 0 \leq x \leq 1$$

- find $u(x)$ given $b(x)$ and boundary conditions;
- our solution strategy is to turn the problem into our favorite form: $Ax = b$;
- the differential operator $-\frac{d^2}{dx^2}$ becomes a square matrix A;
- $u(x)$ becomes a vector of unknowns u.

Discretization of an ODE



- subdivide the interval $0 \leq x \leq 1$ into n cells of size $h = \frac{1}{n}$;
- $u(x)$ is replaced by a vector of unknowns at the grid points $u_i = u(x_i)$;
- at every (interior) grid point approximate the equation by

$$-\frac{1}{h^2}(u_{i-1} - 2u_i + u_{i+1}) = b(x_i);$$

- n equations in the n unknowns $u_1, \ldots, u_n$:

$$\frac{1}{h^2}\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{n-1} \\ u_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix}.$$

Interpretation of Jacobi Method

- consider the 1d Poisson problem on a regular grid, with the tridiagonal A:

$$u_i^{(k+1)} = u_i^{(k)} + \tfrac{1}{2}\big(b_i - \underbrace{(-u_{i-1}^{(k)} + 2u_i^{(k)} - u_{i+1}^{(k)})}_{(Au)_i}\big),$$

$$u_i^{(k+1)} = \frac{1}{2}\left(u_{i-1}^{(k)} + u_{i+1}^{(k)} + b_i\right);$$

- *relaxation* or *smoothing*: at every iteration, the new value at a node is obtained from the average of the values of the two neighbors from the previous iteration;

Interpretation of Gauss-Seidel Method

- consider the 1d Poisson problem on a regular grid, with the tridiagonal A:

$$u_i^{(k+1)} = \frac{1}{2}\left(u_{i-1}^{(k+1)} + u_{i+1}^{(k)} + b_i\right);$$

- *relaxation* or *smoothing*: at every iteration, the new value at a node is obtained from the average of the values of the two neighbors, *using the most up to date values*

Matlab Implementation

```
% given A,b
niter = 5000;
n = size(A, 1);
normb = norm(b);
history = zeros(niter, 1);
M = sparse(diag(diag(A)));  % Jacobi
% M = tril(A);              % Gauss-Seidel
% w = 1.67;
% M = (1-w)/w * diag(diag(A)) + tril(A);   %SOR
for k = 1:niter
r = b - A*x;
history(k) = norm(r)/normb;
x = x + M\r;
end
semilogy(1:niter, history, 'LineWidth', 3)
hold on
```

Convergence of Stationary Methods

stationary method:

$$\mathbf{x}^{(\mathbf{k}+1)} = \mathbf{x}^{(\mathbf{k})} + \mathbf{M}^{-1}\mathbf{r}^{(\mathbf{k})}$$

$\mathbf{x} \leftarrow$ exact solution of $\mathbf{Ax} = \mathbf{b}$
$\mathbf{r}^{(\mathbf{k})} = \mathbf{b} - \mathbf{Ax}^{(\mathbf{k})} \leftarrow$ (known) residual at iteration $\mathbf{k}$
$\mathbf{e}^{(\mathbf{k})} = \mathbf{x} - \mathbf{x}^{(\mathbf{k})} \leftarrow$ (unknown) error at iteration $\mathbf{k}$

goal: discuss the convergence of the method, that is, when does $\mathbf{e}^{(\mathbf{k})} \to \mathbf{0}$ when $\mathbf{k} \to \infty$?

Convergence of Stationary Methods

$$\mathbf{x} = (\mathbf{I} - \mathbf{M}^{-1}\mathbf{A})\mathbf{x} + \mathbf{M}^{-1}\mathbf{b};$$
$$\mathbf{x}^{(\mathbf{k}+1)} = (\mathbf{I} - \mathbf{M}^{-1}\mathbf{A})\mathbf{x}^{(\mathbf{k})} + \mathbf{M}^{-1}\mathbf{b};$$

subtract both equations to get

$$\mathbf{e}^{(\mathbf{k}+1)} = \mathbf{x} - \mathbf{x}^{(\mathbf{k}+1)} = (\mathbf{I} - \mathbf{M}^{-1}\mathbf{A})\mathbf{e}^{(\mathbf{k})} = \mathbf{T}\mathbf{e}^{(\mathbf{k})},$$

in which $\mathbf{T}$ is the iteration matrix.

Convergence of Stationary Methods

The solution of

$$\mathbf{e}^{(\mathbf{k}+\mathbf{1})} = \mathbf{T}\mathbf{e}^{(\mathbf{k})}$$

is

$$\mathbf{e}^{(\mathbf{k})} = \mathbf{T}^{\mathbf{k}}\mathbf{e}^{(\mathbf{0})},$$

in which $\mathbf{e}^{(\mathbf{0})} = \mathbf{x} - \mathbf{x}^{(\mathbf{0})}$ is a fixed initial error. Then, the method converges, that is, $\mathbf{e}^{(\mathbf{k})} \to \mathbf{0}$ when $\mathbf{k} \to \infty$ if and only if

$$\rho(\mathbf{T}) < 1.$$

Moreover, the smaller the $\rho(\mathbf{T})$, the faster the convergence.

Convergence of Stationary Methods

- a sufficient condition for Jacobi and Gauss-Seidel to converge is that $\mathbf{A}$ is strictly diagonally dominant, that is,

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

  in this case, it can be shown that $\rho(\mathbf{I} - \mathbf{M}^{-1}\mathbf{A}) < 1$;
- if $\mathbf{A}$ is SPD, Gauss-Seidel converges; Jacobi may not converge on SPD matrices, but a slight modification to the iteration can guarantee convergence:

$$\rho(\mathbf{I} - \mathbf{w}\mathbf{D}^{-1}\mathbf{A}) < 1 \text{ for a suitable } 0 < w < 2/\lambda_{\max}(\mathbf{D}^{-1}\mathbf{A}).$$

Rate of Convergence

- how fast does the iterative method converge?
- how does $\mathbf{e}^{(\mathbf{k}+\mathbf{1})}$ compare to $\mathbf{e}^{(\mathbf{k})}$?

$$\mathbf{e}^{(\mathbf{k})} = \mathbf{T}^{\mathbf{k}}\mathbf{e}^{(\mathbf{0})} \Rightarrow \quad ||\mathbf{e}^{(\mathbf{k})}|| \approx ||\mathbf{T}^{\mathbf{k}}||\,||\mathbf{e}^{(\mathbf{0})}|| \Rightarrow \quad ||\mathbf{e}^{(\mathbf{k})}|| \approx \rho\left(\mathbf{T}\right)^{\mathbf{k}}||\mathbf{e}^{(\mathbf{0})}||;$$

of we take log on both sides:

$$\log||\mathbf{e}^{(\mathbf{k})}|| \approx k\log\rho\left(\mathbf{T}\right) + \log||\mathbf{e}^{(\mathbf{0})}||,$$

which means that the plot of the log of the error vs. k is a straight line with a negative slope.

Rate of Convergence

- how many iterations until the error norm drops an order of magnitude from $||\mathbf{e}^{(0)}||$?
- that is, how many iterations are needed to reduce the error norm by a factor of 10?

$$||\mathbf{e}^{(\mathbf{k})}|| \approx 0.1||\mathbf{e}^{(0)}|| \approx \rho\left(\mathbf{T}\right)^{\mathbf{k}} ||\mathbf{e}^{(0)}||;$$

take the $\log_{10}$ and solve for k:

$$k = -\frac{1}{\log \rho\left(\mathbf{T}\right)} = \frac{1}{\text{rate}} \quad \Rightarrow \quad \text{rate} = -\log_{10} \rho\left(\mathbf{T}\right).$$

Rate of Convergence and Spectral Radius

- the rate of convergence depends on $\rho(\mathbf{T})$, the spectral radius of the iteration matrix;
- there are reasonable methods we may use to estimate $\lambda_{\max}$ of $\mathbf{I} - \mathbf{M}^{-1}\mathbf{A}$;
- when we know where $\mathbf{A}$ is coming from, we have some *a-priori* estimates: for example, when $\mathbf{A}$ is a discretization of Poisson's equation on a regular grid of spacing h, then

$$\rho(\mathbf{T}) = 1 - ch^2,$$

and as discretization gets more refined, $h \to 0$, convergence becomes very slow.

Termination Criterion: Criterion 1

What are reasonable criteria to terminate the iteration?

A common criterion is to define `tol` and stop when

$$||\mathbf{r}^{(\mathbf{k})}|| \leq \texttt{tol}||\mathbf{b}||;$$

the relative error is then

$$\frac{\left\|\mathbf{x}-\mathbf{x}^{(\mathbf{k})}\right\|}{\|\mathbf{x}\|} \leq \kappa\left(\mathbf{A}\right) \frac{\left\|\mathbf{r}^{(\mathbf{k})}\right\|}{\|\mathbf{b}\|} \leq \kappa\left(\mathbf{A}\right) \texttt{tol};$$

the larger $\kappa\left(\mathbf{A}\right)$, the smaller `tol`.

Termination Criterion: Criterion 2

We can also compare the difference between consecutive iterations: $||\mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{k-1})}||$.

$$||\mathbf{x} - \mathbf{x}^{(\mathbf{k})}|| \leq ||\mathbf{T}||||\mathbf{x} - \mathbf{x}^{(\mathbf{k-1})}|| = ||\mathbf{T}||||\mathbf{x} - \mathbf{x}^{(\mathbf{k})} + \mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{k-1})}||$$
$$\leq ||\mathbf{T}|| \left(||\mathbf{x} - \mathbf{x}^{(\mathbf{k})}|| + ||\mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{k-1})}|| \right)$$

$$\Rightarrow ||\mathbf{e}^{(\mathbf{k})}|| = ||\mathbf{x} - \mathbf{x}^{(\mathbf{k})}|| \leq \frac{||\mathbf{T}||}{1 - ||\mathbf{T}||} ||\mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{k-1})}||;$$

thus, to decide on $||\mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{k-1})}||$, we need to consider $||\mathbf{T}||/(1 - ||\mathbf{T}||)$.

Stationary Methods: Limitation

T, or better **M**, is not updated at each iteration, producing slow convergence.

Updating **M** is the goal of the Conjugate Gradient Method (**A** is SPD) and Krylov Subspace Methods.

Krylov Subspace Methods, such as GMRES and MINRES, can handle other classes of matrices.